

MJ HAPPY HEALTHCARE ERP + CRM

Technical Implementation Document

Developer & QA Reference Document

1. DOCUMENT PURPOSE

This document is prepared for:

- Backend Developers
- Frontend Developers
- Flutter Developers
- QA/Testers
- Project Managers
- DevOps Team

Purpose of this document:

- Define system architecture
 - Define implementation scope
 - Clarify reusable modules vs custom development
 - Define workflows and dependencies
 - Provide development understanding
 - Provide QA validation reference
-

2. PROJECT OVERVIEW

MJ Happy Healthcare requires a cloud-based Pharma Distribution ERP + CRM platform capable of handling:

- Pharma inventory
- Batch/expiry management
- GST billing
- Distribution workflows
- Retailer ordering
- MR field force operations

- GPS tracking
- CRM & lead lifecycle
- Alerts & notifications
- Accounting workflows
- Cloud operations

The system should support:

HEAD OFFICE

↓

National Warehouse / C&F

↓

Super Stockist

↓

Distributor

↓

Retailer / Hospital

↓

Patient

3. BASE ERP FOUNDATION

Selected Foundation

Acnoo Pharmacy SaaS Architecture - [AcnooPharmacy SaaS - Pharmacy Management Software & Flutter App with Admin Panel by Acnoo](#)

Purpose: Use as ERP foundation layer to reduce development timeline.

4. WHAT WILL BE REUSED FROM ERP FOUNDATION

The following modules/features are expected to be reused or partially reused:

4.1 Authentication & User Base

- Login architecture
- User session management
- Password management
- Role foundation

- Basic permissions

4.2 Product Management Base

- Product CRUD
- Categories
- Brands
- Batch structure
- GST structure
- Barcode support
- Basic pricing

4.3 Inventory Base

- Inventory structure
- Stock add/reduce flow
- Purchase entries
- Sales entries
- Batch tracking
- Expiry tracking
- Basic warehouse support

4.4 Billing Base

- Invoice generation
- GST calculations
- Barcode billing
- Sales returns
- Purchase returns

4.5 Reporting Base

- Sales reports
- Purchase reports
- Inventory reports
- Financial summaries

4.6 Mobile Base

- Flutter structure
- API architecture
- Authentication APIs
- Existing mobile UI components

4.7 Admin Dashboard Base

- Dashboard layout
 - Navigation
 - User management
 - Basic analytics
-

5. WHAT WILL BE CUSTOM DEVELOPED

The following modules are NOT sufficiently available in ERP foundation and will require full or major customization.

5.1 DISTRIBUTION HIERARCHY ENGINE

Objective

Create pharma distribution hierarchy.

Required Structure

HO
↓
C&F
↓
Super Stockist
↓
Distributor
↓
Retailer

Functional Requirements

- Multi-level hierarchy management
- Territory allocation
- Hierarchy-based pricing
- Role-wise stock visibility
- Role-wise order visibility
- Parent-child account mapping

Database Requirements

New Tables

- hierarchy_master
- territory_master
- hierarchy_mapping
- distributor_mapping
- retailer_mapping

Backend Requirements

- Role middleware
- Territory access validation
- Hierarchy APIs
- Pricing logic engine

QA Validation

- Verify hierarchy restrictions
 - Verify stock visibility
 - Verify territory mapping
 - Verify user access control
-

5.2 DISTRIBUTOR PORTAL

Objective

Provide dedicated distributor workflow.

Functional Requirements

- Distributor login
- Product catalog
- Order placement
- Outstanding visibility
- Ledger visibility
- Invoice download
- Scheme visibility
- Order history
- Dispatch tracking

Workflow

Distributor Login



Browse Products



Add Products



Place Order



Company Approval



Warehouse Dispatch



Invoice Generated



Delivery Completed

APIs Required

- Distributor dashboard API
- Product listing API
- Order API
- Ledger API
- Invoice API
- Dispatch tracking API

QA Validation

- Verify distributor login
- Verify order placement
- Verify pricing
- Verify invoice generation
- Verify outstanding calculations

5.3 RETAILER ORDERING SYSTEM

Objective

Separate retailer ordering workflow.

Functional Requirements

- Retailer login
- Product browsing
- Offer visibility
- Scheme visibility
- Retail pricing
- Cart management
- Order placement
- Reorder flow
- Invoice access
- Payment tracking

Workflow

Retailer Login

↓

View Products

↓

View Schemes

↓

Place Order

↓

Order Processing

↓

Invoice Generated

↓

Payment Tracking

Database Requirements

- retailer_orders
- retailer_cart
- retailer_pricing
- retailer_schemes

QA Validation

- Verify retailer-specific pricing
- Verify scheme visibility
- Verify reorder functionality
- Verify order history

5.4 CRM MODULE

Objective

Manage pre-sales lifecycle.

CRM Scope

- Lead creation
- Lead assignment
- Follow-up management
- Meeting scheduling
- Activity tracking
- Lead pipeline
- Distributor onboarding
- Doctor CRM
- Hospital CRM

Lead Stages

New Lead

↓

Contacted

↓

Meeting Scheduled

↓

Follow-up

↓

Proposal Shared

↓

Negotiation

↓

Converted

Database Requirements

- crm_leads
- crm_activities
- crm_followups
- crm_meetings
- crm_notes
- crm_pipeline

Backend Requirements

- Lead assignment engine
- Reminder scheduler

- Notification engine
- Activity logger

QA Validation

- Verify lead lifecycle
 - Verify reminders
 - Verify follow-up tracking
 - Verify activity logs
 - Verify conversion workflows
-

5.5 MR (FIELD FORCE) MODULE

Objective

Track MR field operations.

Functional Requirements

- MR login
- Daily schedule
- Doctor visits
- Chemist visits
- Order booking
- Daily reporting
- Meeting tracking
- Tour planning
- Target tracking

Workflow

MR Login

↓

GPS Check-In

↓

Daily Schedule

↓

Doctor Visit

↓

Chemist Visit

↓

Order Booking

↓
Follow-up
↓
Daily Report
↓
GPS Check-Out

APIs Required

- Schedule API
- Visit API
- Reporting API
- Target API
- Order API

QA Validation

- Verify attendance
 - Verify order booking
 - Verify reporting
 - Verify visit logs
-

5.6 GPS TRACKING MODULE

Objective

Track MR location and movement.

Functional Requirements

- GPS attendance
- Check-in/check-out
- Geo-tagged visits
- Daily movement tracking
- Meeting location tracking
- Route history
- Territory validation

Mobile Requirements

- Background location permissions

- GPS tracking integration
- Google Maps integration

Database Requirements

- mr_location_logs
- attendance_logs
- route_tracking
- visit_locations

APIs Required

- Location sync API
- Attendance API
- Route tracking API

QA Validation

- Verify GPS accuracy
 - Verify check-in/check-out
 - Verify geo-tagged visits
 - Verify route tracking
-

5.7 ALERTS & NOTIFICATION ENGINE

Objective

Centralized notification system.

Notification Types

- Payment reminders
- Follow-up reminders
- Meeting reminders
- Expiry alerts
- Reorder alerts
- Dispatch alerts

Channels

- In-app notifications
- Push notifications
- SMS-ready structure
- Email-ready structure

Database Requirements

- notifications
- notification_logs
- scheduled_reminders

QA Validation

- Verify reminder triggers
 - Verify notification delivery
 - Verify scheduling logic
-

5.8 ACCOUNTING ENHANCEMENTS

Scope

- Ledger management
- Outstanding tracking
- Aging reports
- Commission system
- P&L
- Balance sheet
- GST reports

Backend Requirements

- Ledger calculations
- Aging engine
- Commission calculations
- GST calculations

QA Validation

- Verify calculations
- Verify reports
- Verify outstanding logic

5.9 WAREHOUSE & C&F WORKFLOW

Scope

- Inward stock
- Outward stock
- Warehouse transfer
- Dispatch workflows
- LR tracking
- Transport challan

Workflow

Stock Inward

↓

Warehouse Allocation

↓

Distributor Order

↓

Dispatch Challan

↓

LR Assignment

↓

Dispatch

Database Requirements

- warehouse_transfer
- dispatch_master
- transport_details
- lr_tracking

QA Validation

- Verify stock movement
- Verify transfer logs
- Verify dispatch workflows

6. ROLE DEFINITIONS

Roles

- Super Admin
- Management
- Warehouse Admin
- C&F User
- Super Stockist
- Distributor
- Retailer
- MR
- Accountant

Access Rules

Each role should:

- See only assigned modules
 - Access assigned territory
 - View assigned pricing
 - Access assigned reports
-

7. API ARCHITECTURE

API Type

REST APIs

Authentication

- JWT/Auth token-based

Modules

- Authentication APIs
- Product APIs
- Inventory APIs
- CRM APIs
- Order APIs
- MR APIs
- GPS APIs
- Notification APIs

- Accounting APIs

QA Validation

- Verify API security
 - Verify token expiry
 - Verify unauthorized access restrictions
-

8. MOBILE APPLICATION DETAILS

Technology

Flutter

Mobile Users

- MR
- Distributor
- Retailer
- Management

Mobile Features

- Dashboard
- Ordering
- Visit tracking
- GPS attendance
- Notifications
- Reporting

QA Validation

- Verify responsiveness
 - Verify API sync
 - Verify GPS permissions
 - Verify offline handling
-

9. CLOUD DEPLOYMENT

Hosting

Cloud VPS / AWS / DigitalOcean

Deployment Scope

- Production server setup
- SSL setup
- Database setup
- Backup configuration
- Environment configuration

DevOps Requirements

- Git repository
 - CI/CD preferred
 - Staging environment
 - Production environment
-

10. DATABASE GUIDELINES

Important Rules

- Avoid hardcoded values
- Use relational mapping properly
- Use indexing where required
- Maintain audit logs
- Use soft delete where needed

Critical Tables

- users
- roles
- hierarchy_master
- crm_leads
- inventory_batches
- retailer_orders
- distributor_orders
- attendance_logs
- notification_logs

11. QA TESTING CHECKLIST

Functional Testing

- Authentication
- Role access
- Billing
- Inventory
- CRM
- GPS tracking
- Notifications
- Orders
- Dispatch

Performance Testing

- API response time
- Dashboard loading
- Bulk order handling
- Inventory reports

Security Testing

- Unauthorized access
- API validation
- SQL injection checks
- Session handling

Mobile Testing

- Android compatibility
 - GPS permissions
 - Notification delivery
 - App crashes
-

12. FUTURE-READY STRUCTURE

The system architecture should support future integration for:

- Export workflows
- Multi-currency
- Manufacturing ERP
- AI analytics
- Franchise/PCD
- Advanced BI dashboards
- WhatsApp automation
- E-commerce integrations

Functional Requirement Document (FRD)

Module Set 1

Modules Covered:

1. Master Admin Panel
2. Product Management
3. Inventory & Warehouse Management

MODULE 1 — MASTER ADMIN PANEL

1.1 MODULE OVERVIEW

Module Name

Master Admin Panel

Objective

Provide centralized operational and management control for the complete Pharma ERP + CRM ecosystem.

This module acts as the primary monitoring and control center for:

- Sales
- Inventory
- Distribution
- CRM

- MR activities
 - Accounting
 - Notifications
 - Business analytics
-

1.2 WHO WILL USE THIS MODULE

User Role	Purpose
Super Admin	Complete platform control
Directors	Business monitoring
Management Team	Operations monitoring
Finance Team	Payment & accounting visibility
Operations Team	Dispatch & inventory visibility

1.3 PRE-CONDITIONS

Before using this module:

- User account should exist.
 - User role & permissions should be configured.
 - Product master should exist.
 - Inventory setup should be completed.
 - Distributor hierarchy should be configured.
 - Warehouses/C&F should be configured.
 - CRM setup should be completed.
 - Notification engine should be active.
-

1.4 POST-CONDITIONS

After successful usage:

- Dashboard metrics should refresh.
- Activity logs should be maintained.
- Reports should be generated.

- Notifications should be triggered.
 - Real-time monitoring should be available.
-

1.5 FUNCTIONAL REQUIREMENTS

FR-1: Secure Admin Login

User Story

As a Super Admin, I want secure access to the admin panel so that only authorized users can access sensitive operational and financial data.

Flow of Work

1. User opens admin portal.
 2. User enters:
 - Username/email
 - Password
 3. System validates credentials.
 4. System validates role permissions.
 5. JWT/session token generated.
 6. Dashboard loads according to permissions.
-

Error Handling

Scenario	Handling
Invalid credentials	Show validation message
Inactive user	Restrict access
Unauthorized role	Access denied
Multiple failed attempts	Temporary account lock
Session expired	Redirect to login

Acceptance Criteria

- Only authorized users can login.
- Session timeout should work.
- Role-based menus should display correctly.
- Login activity should be logged.
- Unauthorized users should not access restricted modules.

Related Modules

- Authentication Module
- Role Management
- Notifications
- Audit Logs

Impact on Other Screens

- Dashboard visibility changes based on role.
- Menu access changes dynamically.
- Reports access changes dynamically.

Module Communication

Source Module	Target Module	Communication
Authentication	Admin Panel	Login validation
Role Management	Admin Panel	Permission access
Audit Logs	Admin Panel	User activity tracking

FR-2: All India Sales Dashboard

User Story

As management, I want to monitor overall business performance so that I can make operational and financial decisions.

Flow of Work

1. User logs in.
 2. Dashboard APIs called.
 3. System fetches:
 - Sales
 - Orders
 - Inventory
 - CRM data
 - Outstanding amounts
 4. KPI aggregation executed.
 5. Dashboard widgets rendered.
-

Dashboard Components

- Total Sales
 - State-wise Sales
 - Outstanding Amount
 - Top Products
 - Pending Dispatch
 - Low Stock Alerts
 - Near Expiry Alerts
 - Distributor Performance
 - Lead Funnel Summary
 - MR Activity Summary
-

Error Handling

Scenario	Handling
API failure	Retry mechanism
Empty data	Empty state message
Slow response	Timeout fallback
Partial data sync	Last sync timestamp display

Acceptance Criteria

- Dashboard should load correctly.
 - KPIs should reflect real-time data.
 - Filters should work properly.
 - Graphs should load correctly.
 - Dashboard should support mobile responsiveness.
-

Related Modules

- Billing
 - Inventory
 - CRM
 - Accounting
 - Distributor Orders
 - Notifications
 - MR Module
-

Impact on Other Screens

- KPI click redirects to detailed reports.
 - Alerts redirect to operational screens.
 - Dashboard filters affect report screens.
-

Module Communication

Source Module	Target Module	Communication
Billing	Dashboard	Sales data
Inventory	Dashboard	Stock data
CRM	Dashboard	Lead statistics
Accounting	Dashboard	Outstanding values
MR Module	Dashboard	Field activity summary

FR-3: Pending Payment Monitoring

User Story

As finance management, I want to monitor pending payments so that collections and outstanding management can be controlled.

Flow of Work

1. Invoice generated.
 2. Outstanding entry created.
 3. Aging calculation executed.
 4. Due date monitored.
 5. Reminder notifications triggered.
 6. Payment updates reflected.
-

Error Handling

Scenario	Handling
Invoice mismatch	Flag reconciliation issue
Duplicate invoice	Restrict duplicate processing
Missing customer mapping	Validation error
Payment sync issue	Pending reconciliation status

Acceptance Criteria

- Outstanding calculations should be accurate.
 - Aging reports should generate correctly.
 - Reminders should trigger automatically.
 - Payment updates should reflect in real-time.
-

Related Modules

- Billing
 - Accounting
 - Notifications
 - Distributor Portal
 - Retailer Orders
-

Impact on Other Screens

- Outstanding visible in distributor portal.
 - Alerts visible in dashboard.
 - Collection reports updated.
-

Module Communication

Source Module	Target Module	Communication
Billing	Accounting	Invoice sync
Accounting	Notifications	Payment reminders
Distributor Portal	Accounting	Payment updates

1.6 QA VALIDATION CHECKLIST

- Verify role-based login.
 - Verify dashboard KPIs.
 - Verify dashboard filters.
 - Verify outstanding calculations.
 - Verify payment reminder triggers.
 - Verify report redirections.
 - Verify session timeout.
-
-

MODULE 2 — PRODUCT MANAGEMENT

2.1 MODULE OVERVIEW

Module Name

Product Management

Objective

Manage complete pharma product lifecycle including:

- Product creation
 - Batch management
 - Expiry tracking
 - Pricing
 - GST
 - Barcode
 - Schemes
-

2.2 WHO WILL USE THIS MODULE

User Role	Purpose
Admin	Product configuration
Inventory Team	Batch & stock management
Warehouse Team	Batch inward/outward
Billing Team	Billing product usage
Management	Pricing & scheme control

2.3 PRE-CONDITIONS

- Brand master should exist.
- Category master should exist.
- GST setup should exist.
- User permissions should be configured.
- Inventory module should be active.

2.4 POST-CONDITIONS

- Product available in inventory.
 - Product available in billing.
 - Product available in ordering systems.
 - Product visible in reports.
 - Product available for schemes.
-

2.5 FUNCTIONAL REQUIREMENTS

FR-1: Product Creation

User Story

As admin, I want to create pharma products so that inventory, billing, ordering, and distribution operations can be managed.

Flow of Work

1. User opens product management screen.
2. User clicks Add Product.
3. User enters:
 - Product Name
 - Brand
 - Category
 - Composition
 - Packing Type
 - GST
 - MRP
 - PTR
 - PTS
 - PTD
 - Barcode
4. Product saved.

5. Product synced across modules.
-

Error Handling

Scenario	Handling
Duplicate barcode	Restrict save
Missing mandatory fields	Validation error
Invalid GST	Validation error
Duplicate product code	Restrict save

Acceptance Criteria

- Product should save successfully.
 - Product should appear in inventory.
 - Product should appear in billing.
 - Product should appear in ordering modules.
 - Barcode should remain unique.
-

Related Modules

- Inventory
 - Billing
 - Distributor Ordering
 - Retailer Ordering
 - CRM Promotions
 - Reporting
-

Impact on Other Screens

- Product visible in billing screen.
 - Product visible in retailer catalog.
 - Product visible in distributor ordering.
 - Product visible in analytics.
-

Module Communication

Source Module	Target Module	Communication
Product Module	Inventory	Product sync
Product Module	Billing	Product listing
Product Module	Ordering	Product availability
Product Module	CRM	Promotional mapping

FR-2: Batch & Expiry Management

User Story

As inventory team, I want batch-wise inventory tracking so that expiry and compliance can be managed accurately.

Flow of Work

1. Stock inward initiated.
 2. Batch created.
 3. Expiry date assigned.
 4. Batch stock stored separately.
 5. Expiry monitoring initiated.
 6. Alerts triggered for near expiry.
-

Error Handling

Scenario	Handling
Duplicate batch number	Warning/validation
Invalid expiry date	Restrict save
Expired batch inward	Restrict processing
Negative stock	Prevent transaction

Acceptance Criteria

- Batch inventory should track separately.
 - Expiry alerts should trigger correctly.
 - Billing should use valid batches.
 - Expired products should not sell.
-

Related Modules

- Inventory
 - Billing
 - Alerts
 - Warehouse
 - Dashboard
-

Impact on Other Screens

- Expiry alerts visible in dashboard.
 - Batch selection visible during billing.
 - Expiry reports updated.
-

Module Communication

Source Module	Target Module	Communication
Inventory	Alerts	Expiry alerts
Batch Module	Billing	Batch availability
Warehouse	Inventory	Batch stock sync

FR-3: Scheme Management

User Story

As management, I want to configure product schemes so that promotional offers can be provided to distributors and retailers.

Flow of Work

1. Admin opens scheme module.
 2. Product selected.
 3. Scheme rules configured.
 4. Validity dates configured.
 5. Scheme activated.
 6. Order modules apply scheme automatically.
-

Error Handling

Scenario	Handling
Invalid scheme dates	Validation error
Duplicate active scheme	Warning
Product not mapped	Restrict activation

Acceptance Criteria

- Scheme should apply automatically.
- Expired schemes should deactivate automatically.
- Role-based scheme visibility should work.
- Discounts should reflect in invoice.

MODULE 4 — DISTRIBUTOR PORTAL

4.1 MODULE OVERVIEW

Module Name

Distributor Portal

Objective

Provide distributors and stockists with a dedicated self-service ordering and account management platform.

This module will manage:

- Product ordering
 - Outstanding visibility
 - Invoice access
 - Ledger visibility
 - Dispatch tracking
 - Scheme visibility
 - Order history
-

4.2 WHO WILL USE THIS MODULE

User Role	Purpose
Distributor	Product ordering
Super Stockist	Bulk ordering
Sales Team	Distributor monitoring
Accounts Team	Payment tracking
Management	Distributor performance visibility

4.3 PRE-CONDITIONS

- Distributor account should exist.
- Distributor should be mapped to territory.
- Pricing should be configured.
- Product catalog should be active.
- Outstanding logic should be configured.
- Warehouse stock should be available.

4.4 POST-CONDITIONS

- Orders created successfully.
 - Outstanding updated.
 - Dispatch workflow initiated.
 - Notifications triggered.
 - Order history maintained.
-

4.5 FUNCTIONAL REQUIREMENTS

FR-1: Distributor Login & Dashboard

User Story

As a distributor, I want to securely login and access my operational dashboard so that I can manage orders, payments, and inventory visibility.

Flow of Work

1. Distributor opens portal/app.
 2. Distributor enters credentials.
 3. Authentication validated.
 4. Territory & account status checked.
 5. Dashboard loads.
-

Dashboard Components

- Outstanding Summary
- Recent Orders
- Pending Dispatch
- Active Schemes
- Payment Due Alerts

- Order Status
 - Product Availability
-

Error Handling

Scenario	Handling
Invalid login	Validation error
Inactive distributor	Restrict login
Territory not mapped	Access restriction
Account blocked	Display support message

Acceptance Criteria

- Distributor should login successfully.
 - Dashboard data should load correctly.
 - Outstanding values should be accurate.
 - Order status should reflect correctly.
-

Related Modules

- Authentication
 - Inventory
 - Billing
 - Accounting
 - Notifications
-

Impact on Other Screens

- Outstanding visible in finance reports.
 - Order visibility reflected in dispatch module.
 - Scheme visibility reflected from product module.
-

Module Communication

Source Module	Target Module	Communication
Authentication	Distributor Portal	Login validation
Inventory	Distributor Portal	Stock availability
Accounting	Distributor Portal	Outstanding sync
Notifications	Distributor Portal	Alerts display

FR-2: Distributor Order Placement

User Story

As a distributor, I want to place product orders digitally so that procurement becomes faster and centralized.

Flow of Work

1. Distributor browses products.
 2. Product pricing fetched.
 3. Schemes applied.
 4. Cart updated.
 5. Order submitted.
 6. Company approval workflow initiated.
 7. Dispatch workflow triggered.
-

Error Handling

Scenario	Handling
Product out of stock	Restrict ordering
Invalid pricing	Refresh pricing
Expired scheme	Remove scheme

Order duplication Duplicate warning

Acceptance Criteria

- Orders should save successfully.
 - Correct pricing should apply.
 - Schemes should calculate automatically.
 - Orders should reflect in dispatch queue.
-

Related Modules

- Product Management
 - Inventory
 - Billing
 - Warehouse
 - Notifications
-

Impact on Other Screens

- Inventory reserved after approval.
 - Dispatch queue updated.
 - Order visible in admin reports.
-

Module Communication

Source Module	Target Module	Communication
Product Module	Orders	Product pricing
Inventory	Orders	Stock validation
Orders	Warehouse	Dispatch request
Orders	Billing	Invoice generation

FR-3: Invoice, Ledger & Outstanding Visibility

User Story

As a distributor, I want to access invoices and outstanding details so that I can manage payments and accounting transparency.

Flow of Work

1. Distributor opens finance section.
 2. System fetches:
 - Invoices
 - Outstanding
 - Ledger
 - Payment history
 3. Distributor downloads invoice if required.
 4. Payment reminders displayed.
-

Error Handling

Scenario	Handling
Missing invoice	Show unavailable status
Ledger mismatch	Flag reconciliation issue
Download failure	Retry option

Acceptance Criteria

- Ledger should match accounting module.
 - Outstanding should calculate correctly.
 - Invoice download should work.
 - Payment history should reflect correctly.
-

Related Modules

- Accounting
 - Billing
 - Notifications
 - Reports
-

Impact on Other Screens

- Outstanding visible in admin dashboard.
 - Payment updates reflected in finance reports.
-

Module Communication

Source Module	Target Module	Communication
Billing	Distributor Portal	Invoice sync
Accounting	Distributor Portal	Ledger sync
Notifications	Distributor Portal	Payment reminders

4.6 QA VALIDATION CHECKLIST

- Verify distributor login.
 - Verify order placement.
 - Verify scheme calculations.
 - Verify invoice download.
 - Verify ledger calculations.
 - Verify outstanding sync.
 - Verify dispatch visibility.
-
-

MODULE 5 — RETAILER ORDERING SYSTEM

5.1 MODULE OVERVIEW

Module Name

Retailer Ordering System

Objective

Provide medical stores and hospitals with a dedicated digital ordering platform.

This module manages:

- Product browsing
 - Retail ordering
 - Offer visibility
 - Scheme visibility
 - Payment visibility
 - Reordering
-

5.2 WHO WILL USE THIS MODULE

User Role	Purpose
Retail Medical Store	Medicine ordering
Hospitals	Bulk medicine ordering
Sales Team	Retail order visibility
Management	Retail analytics

5.3 PRE-CONDITIONS

- Retailer account should exist.
 - Product catalog should exist.
 - Retail pricing configured.
 - Distributor mapping completed.
 - Active schemes configured.
-

5.4 POST-CONDITIONS

- Orders created.
 - Retailer history updated.
 - Notifications triggered.
 - Payment tracking updated.
-

5.5 FUNCTIONAL REQUIREMENTS

FR-1: Retail Product Browsing

User Story

As a retailer, I want to browse available products and offers so that I can place purchase orders efficiently.

Flow of Work

1. Retailer logs in.
 2. Product catalog loads.
 3. Pricing & schemes displayed.
 4. Search/filter functionality available.
 5. Retailer selects products.
-

Error Handling

Scenario	Handling
Product unavailable	Disable ordering
Pricing mismatch	Refresh pricing
Expired offer	Remove offer display

Acceptance Criteria

- Products should load correctly.
 - Retail pricing should display correctly.
 - Schemes should display correctly.
 - Product search/filter should work.
-

Related Modules

- Product Management
 - Inventory
 - Schemes
 - Notifications
-

Impact on Other Screens

- Product analytics updated.
 - Retail demand reflected in reports.
-

Module Communication

Source Module	Target Module	Communication
Product Module	Retailer App	Product catalog
Scheme Module	Retailer App	Offer visibility
Inventory	Retailer App	Stock visibility

FR-2: Retailer Order Placement

User Story

As a retailer, I want to place medicine orders digitally so that procurement becomes easier and faster.

Flow of Work

1. Retailer adds products to cart.
 2. Quantity validation happens.
 3. Pricing & schemes applied.
 4. Order submitted.
 5. Distributor/company receives order.
 6. Dispatch initiated.
-

Error Handling

Scenario	Handling
Insufficient stock	Restrict quantity
Invalid pricing	Refresh pricing
Cart session expired	Restore cart
Duplicate order	Warning message

Acceptance Criteria

- Orders should place successfully.
 - Correct pricing should apply.
 - Correct schemes should apply.
 - Order should reflect in dispatch workflow.
-

Related Modules

- Inventory
 - Billing
 - Distributor Portal
 - Notifications
 - Warehouse
-

Impact on Other Screens

- Dispatch queue updated.
 - Inventory reduced after approval.
 - Retail sales analytics updated.
-

Module Communication

Source Module	Target Module	Communication
Retail Orders	Warehouse	Dispatch request
Inventory	Retail Orders	Stock validation
Orders	Billing	Invoice generation
Notifications	Retailer App	Order updates

FR-3: Reorder & Payment Tracking

User Story

As a retailer, I want reorder and payment visibility so that repeat purchasing and payment tracking become easier.

Flow of Work

1. Retailer opens order history.
2. Previous orders displayed.
3. Reorder option available.
4. Outstanding displayed.

5. Payment reminders shown.
-

Error Handling

Scenario	Handling
Old product unavailable	Suggest alternatives
Invoice missing	Show unavailable status
Payment sync delay	Last updated timestamp

Acceptance Criteria

- Reorder should work correctly.
 - Outstanding should reflect correctly.
 - Payment reminders should trigger.
 - Order history should maintain correctly.
-

Related Modules

- Accounting
 - Billing
 - Notifications
 - Inventory
-

Impact on Other Screens

- Outstanding visible in admin dashboard.
 - Retail demand analytics updated.
-

Module Communication

Source Module	Target Module	Communication
---------------	---------------	---------------

Accounting	Retailer App	Outstanding sync
Billing	Retailer App	Invoice sync
Notifications	Retailer App	Reminder alerts

5.6 QA VALIDATION CHECKLIST

- Verify product visibility.
 - Verify retail pricing.
 - Verify order placement.
 - Verify reorder functionality.
 - Verify payment tracking.
 - Verify offer visibility.
 - Verify order history.
-
-

MODULE 6 — WHOLESALE BILLING SYSTEM

6.1 MODULE OVERVIEW

Module Name

Wholesale Billing System

Objective

Manage pharma billing operations with GST compliance and inventory synchronization.

This module handles:

- GST billing
- Barcode billing
- E-invoice support

- Credit notes
 - Sales returns
 - Expiry returns
 - Multi-rate billing
-

6.2 WHO WILL USE THIS MODULE

User Role	Purpose
Billing Team	Invoice generation
Accounts Team	Financial tracking
Warehouse Team	Stock sync
Management	Sales monitoring

6.3 PRE-CONDITIONS

- Product should exist.
 - Batch should exist.
 - GST settings configured.
 - Customer/distributor should exist.
 - Inventory available.
-

6.4 POST-CONDITIONS

- Invoice generated.
 - Stock reduced.
 - Accounting updated.
 - Outstanding created.
 - Reports updated.
-

6.5 FUNCTIONAL REQUIREMENTS

FR-1: GST Invoice Generation

User Story

As billing operator, I want to generate GST-compliant invoices so that billing operations remain compliant and accurate.

Flow of Work

1. Customer selected.
 2. Products scanned/selected.
 3. Batch selected.
 4. GST calculated.
 5. Discounts/schemes applied.
 6. Invoice generated.
 7. Stock reduced.
-

Error Handling

Scenario	Handling
Invalid GST	Validation error
Expired batch	Restrict billing
Insufficient stock	Restrict billing
Duplicate invoice	Prevent duplication

Acceptance Criteria

- GST should calculate correctly.
 - Invoice should generate successfully.
 - Inventory should reduce correctly.
 - Ledger should update correctly.
-

Related Modules

- Inventory
 - Accounting
 - Product Management
 - Notifications
-

Impact on Other Screens

- Sales reports updated.
 - Outstanding reports updated.
 - Inventory reports updated.
-

Module Communication

Source Module	Target Module	Communication
Billing	Inventory	Stock reduction
Billing	Accounting	Ledger update
Billing	Dashboard	Sales summary

FR-2: Barcode Billing

User Story

As billing operator, I want barcode-based billing so that invoice processing becomes faster.

Flow of Work

1. Barcode scanned.
2. Product fetched.
3. Batch validated.
4. Quantity updated.
5. Invoice generated.

Error Handling

Scenario	Handling
Invalid barcode	Validation error
Product unavailable	Restrict billing
Barcode mismatch	Warning message

Acceptance Criteria

- Barcode should fetch correct product.
 - Billing speed should improve.
 - Batch validation should work.
-

Related Modules

- Product Management
 - Inventory
 - Accounting
-

Impact on Other Screens

- Billing speed metrics updated.
 - Inventory movement updated.
-

Module Communication

Source Module	Target Module	Communication
Product Module	Billing	Barcode mapping
Inventory	Billing	Batch validation

FR-3: Sales Return & Expiry Return

User Story

As billing/accounts team, I want to process sales returns and expiry returns so that inventory and financial corrections can be handled.

Flow of Work

1. Return initiated.
 2. Original invoice fetched.
 3. Batch validated.
 4. Return quantity validated.
 5. Stock adjusted.
 6. Credit note generated.
-

Error Handling

Scenario	Handling
Invalid invoice	Restrict processing
Return exceeds sold quantity	Validation error
Batch mismatch	Restrict return

Acceptance Criteria

- Returns should process correctly.
 - Inventory should update correctly.
 - Credit note should generate.
 - Accounting should update.
-

Related Modules

- Inventory
 - Accounting
 - Product Management
 - Reports
-

Impact on Other Screens

- Stock reports updated.
 - Return analytics updated.
 - Outstanding adjusted.
-

Module Communication

Source Module	Target Module	Communication
Returns	Inventory	Stock adjustment
Returns	Accounting	Credit note sync
Returns	Reports	Return analytics

6.6 QA VALIDATION CHECKLIST

- Verify GST calculations.
- Verify barcode billing.
- Verify invoice generation.
- Verify stock reduction.
- Verify sales returns.
- Verify expiry returns.
- Verify credit notes.
- Verify accounting sync.

MODULE 7 — MR (MEDICAL REPRESENTATIVE) MODULE

7.1 MODULE OVERVIEW

Module Name

MR (Medical Representative) Module

Objective

Manage complete field-force operations for Medical Representatives including:

- Daily scheduling
- Doctor visits
- Chemist visits
- Order booking
- Daily reporting
- Tour planning
- Target tracking
- Meeting tracking
- Activity monitoring

This module acts as the core Field Force Automation (FFA) system.

7.2 WHO WILL USE THIS MODULE

User Role	Purpose
Medical Representatives	Daily field operations
Sales Managers	MR monitoring
Management	Performance monitoring
CRM Team	Follow-up visibility
Operations Team	Order coordination

7.3 PRE-CONDITIONS

- MR account should exist.
 - Territory should be assigned.
 - Doctor/Chemist master should exist.
 - GPS permissions should be enabled.
 - CRM module should be active.
 - Product catalog should be available.
-

7.4 POST-CONDITIONS

- MR activities logged.
 - Orders synced.
 - GPS activities tracked.
 - Reports generated.
 - CRM activities updated.
 - Notifications triggered.
-

7.5 FUNCTIONAL REQUIREMENTS

FR-1: MR Login & Daily Dashboard

User Story

As an MR, I want to access my daily dashboard so that I can manage my visits, schedules, targets, and reporting efficiently.

Flow of Work

1. MR opens mobile app.
2. MR enters credentials.

3. Authentication validated.
 4. Territory validation executed.
 5. Dashboard loads.
-

Dashboard Components

- Daily Schedule
 - Pending Visits
 - Doctor Meetings
 - Chemist Visits
 - Daily Targets
 - Order Summary
 - Notifications
 - Attendance Status
-

Error Handling

Scenario	Handling
Invalid login	Validation error
Territory inactive	Restrict access
GPS disabled	Show GPS warning
Internet unavailable	Offline mode warning

Acceptance Criteria

- MR should login successfully.
 - Dashboard should load correctly.
 - Daily schedule should display correctly.
 - Notifications should display correctly.
-

Related Modules

- Authentication
- GPS Tracking
- CRM
- Notifications

- Orders
-

Impact on Other Screens

- Attendance visible in management dashboard.
 - MR activities reflected in reports.
 - CRM follow-ups synced.
-

Module Communication

Source Module	Target Module	Communication
Authentication	MR App	Login validation
CRM	MR Dashboard	Follow-up sync
Notifications	MR Dashboard	Alerts sync
GPS	MR Dashboard	Attendance status

FR-2: Doctor Visit Management

User Story

As an MR, I want to record doctor visits so that my field activities and follow-ups can be tracked.

Flow of Work

1. MR selects doctor.
2. Visit check-in initiated.
3. GPS validation executed.
4. Meeting notes entered.
5. Follow-up added.
6. Visit submitted.

Captured Data

- Doctor Name
 - Visit Time
 - GPS Location
 - Discussion Notes
 - Product Promotion
 - Follow-up Date
 - Sample Distribution
-

Error Handling

Scenario	Handling
GPS unavailable	Restrict check-in
Doctor not mapped	Validation error
Duplicate visit	Warning message
Missing notes	Validation prompt

Acceptance Criteria

- Visit should save successfully.
 - GPS coordinates should capture.
 - Follow-up should sync to CRM.
 - Activity logs should update.
-

Related Modules

- GPS Tracking
 - CRM
 - Notifications
 - Reporting
-

Impact on Other Screens

- Doctor activity visible in CRM.
 - MR performance reports updated.
 - Follow-up reminders triggered.
-

Module Communication

Source Module	Target Module	Communication
Doctor Visit	CRM	Follow-up sync
GPS	Doctor Visit	Location validation
Notifications	MR App	Reminder alerts

FR-3: Chemist Visit & Order Booking

User Story

As an MR, I want to book orders during chemist visits so that product ordering becomes faster and more accurate.

Flow of Work

1. MR selects chemist.
 2. Visit check-in executed.
 3. Product catalog loaded.
 4. MR selects products.
 5. Quantity entered.
 6. Scheme/pricing applied.
 7. Order submitted.
-

Error Handling

Scenario	Handling
----------	----------

Product unavailable	Restrict order
Invalid quantity	Validation error
Pricing mismatch	Refresh pricing
Duplicate order	Warning message

Acceptance Criteria

- Orders should save successfully.
 - Pricing should apply correctly.
 - Scheme calculations should work.
 - Orders should sync to distributor workflow.
-

Related Modules

- Inventory
 - Distributor Orders
 - Pricing
 - Schemes
 - Billing
-

Impact on Other Screens

- Orders visible in distributor queue.
 - Inventory demand updated.
 - MR sales reports updated.
-

Module Communication

Source Module	Target Module	Communication
MR Orders	Distributor Orders	Order sync
Pricing Module	MR Orders	Price calculation

FR-4: Daily Reporting & Target Tracking

User Story

As management, I want MR daily reports and target tracking so that field performance can be monitored effectively.

Flow of Work

1. MR opens reporting screen.
 2. Daily activities displayed.
 3. Target achievement calculated.
 4. MR submits report.
 5. Manager review available.
-

Report Components

- Doctor Visits
 - Chemist Visits
 - Orders Booked
 - Sales Value
 - Pending Follow-ups
 - Tour Summary
-

Error Handling

Scenario	Handling
Incomplete report	Validation error
Missing attendance	Restrict submission
Sync failure	Retry mechanism

Acceptance Criteria

- Daily report should submit correctly.
 - Targets should calculate correctly.
 - Reports should reflect in dashboard.
-

Related Modules

- CRM
 - Dashboard
 - Reporting
 - GPS Tracking
-

Impact on Other Screens

- Management dashboard updated.
 - Performance analytics updated.
 - Incentive calculations updated.
-

Module Communication

Source Module	Target Module	Communication
MR Reporting	Dashboard	Performance sync
CRM	Reporting	Follow-up summary
GPS	Reporting	Attendance sync

7.6 QA VALIDATION CHECKLIST

- Verify MR login.
- Verify doctor visits.
- Verify chemist visits.
- Verify order booking.
- Verify GPS validation.
- Verify target tracking.

- Verify reporting submission.
 - Verify CRM sync.
-
-

MODULE 8 — GPS & LOCATION TRACKING MODULE

8.1 MODULE OVERVIEW

Module Name

GPS & Location Tracking Module

Objective

Track real-time MR movement, attendance, geo-tagged visits, and territory activities.

This module manages:

- GPS attendance
 - Check-in/check-out
 - Geo-tagged visits
 - Route tracking
 - Territory tracking
 - Meeting location validation
-

8.2 WHO WILL USE THIS MODULE

User Role	Purpose
MR Team	Attendance & visit tracking
Sales Managers	Team monitoring
Management	Live movement visibility

8.3 PRE-CONDITIONS

- MR account should exist.
 - GPS permissions enabled.
 - Internet connectivity available.
 - Territory assigned.
 - Mobile app installed.
-

8.4 POST-CONDITIONS

- Attendance logged.
 - Routes tracked.
 - Geo-tagged activities saved.
 - Territory monitoring enabled.
 - Activity logs updated.
-

8.5 FUNCTIONAL REQUIREMENTS

FR-1: GPS Attendance

User Story

As an MR, I want GPS-based attendance so that my field attendance can be validated accurately.

Flow of Work

1. MR opens attendance screen.
2. GPS permission validated.

3. Current location captured.
 4. Attendance marked.
 5. Timestamp stored.
-

Error Handling

Scenario	Handling
GPS disabled	Restrict attendance
Location unavailable	Retry location fetch
Mock GPS detected	Restrict attendance
Duplicate attendance	Warning message

Acceptance Criteria

- Attendance should capture correctly.
 - GPS coordinates should store accurately.
 - Duplicate attendance should restrict.
-

Related Modules

- MR Module
 - Dashboard
 - Reporting
 - HR
-

Impact on Other Screens

- Attendance visible in dashboard.
 - Daily reporting linked.
 - Payroll/incentive calculations linked.
-

Module Communication

Source Module	Target Module	Communication
GPS Module	HR	Attendance sync
GPS Module	Dashboard	Live attendance
GPS Module	Reporting	Attendance logs

FR-2: Geo-Tagged Doctor/Chemist Visits

User Story

As management, I want geo-tagged visit validation so that fake field reporting can be minimized.

Flow of Work

1. MR initiates visit.
 2. GPS captured.
 3. Doctor/Chemist location matched.
 4. Geo-validation executed.
 5. Visit approved.
-

Error Handling

Scenario	Handling
Location mismatch	Flag suspicious activity
GPS unavailable	Restrict submission
Fake GPS detected	Block activity

Acceptance Criteria

- Geo-tagging should capture correctly.
- Visit validation should work.

- Fake location attempts should restrict.
-

Related Modules

- MR Module
 - CRM
 - Notifications
 - Reporting
-

Impact on Other Screens

- Visit reports updated.
 - Performance reports updated.
 - Territory analytics updated.
-

Module Communication

Source Module	Target Module	Communication
GPS	CRM	Visit validation
GPS	Reporting	Location analytics
GPS	Dashboard	Territory visibility

FR-3: Route Tracking & Territory Monitoring

User Story

As sales management, I want route and territory tracking so that field-force productivity can be monitored.

Flow of Work

1. MR starts day.
 2. GPS tracking initiated.
 3. Route points stored periodically.
 4. Territory validation executed.
 5. Route summary generated.
-

Error Handling

Scenario	Handling
GPS interruption	Retry tracking
Network failure	Offline sync
Territory violation	Alert generation

Acceptance Criteria

- Route history should store correctly.
 - Territory tracking should work.
 - Offline sync should function properly.
-

Related Modules

- Dashboard
 - Reporting
 - Notifications
 - CRM
-

Impact on Other Screens

- Route analytics updated.
 - Territory reports updated.
 - Manager dashboard updated.
-

Module Communication

Source Module	Target Module	Communication
GPS	Dashboard	Live movement
GPS	Reporting	Route analytics
GPS	Notifications	Territory alerts

8.6 QA VALIDATION CHECKLIST

- Verify GPS attendance.
 - Verify geo-tagging.
 - Verify route tracking.
 - Verify offline sync.
 - Verify territory validation.
 - Verify fake GPS restrictions.
 - Verify attendance reporting.
-
-

MODULE 9 — CRM & LEAD LIFECYCLE MODULE

9.1 MODULE OVERVIEW

Module Name

CRM & Lead Lifecycle Module

Objective

Manage complete lead lifecycle from inquiry to conversion for:

- Distributors

- Retailers
- Hospitals
- Doctors
- Franchise/PCD opportunities

This module manages:

- Lead creation
 - Lead assignment
 - Follow-ups
 - Meetings
 - Activity tracking
 - Lead conversion
 - Sales pipeline
-

9.2 WHO WILL USE THIS MODULE

User Role	Purpose
Sales Team	Lead handling
MR Team	Field follow-ups
CRM Team	Lead monitoring
Management	Sales pipeline visibility
Distributor Onboarding Team	Partner onboarding

9.3 PRE-CONDITIONS

- CRM setup completed.
 - User permissions configured.
 - Territory mapping configured.
 - Notification engine active.
 - Customer master active.
-

9.4 POST-CONDITIONS

- Lead activities logged.

- Follow-ups scheduled.
 - Notifications triggered.
 - Conversion tracking updated.
 - Pipeline reports updated.
-

9.5 FUNCTIONAL REQUIREMENTS

FR-1: Lead Creation & Assignment

User Story

As sales/CRM team, I want to create and assign leads so that opportunities can be tracked properly.

Flow of Work

1. Lead created.
 2. Lead type selected.
 3. Territory assigned.
 4. Lead owner assigned.
 5. Follow-up scheduled.
 6. Notification triggered.
-

Lead Types

- Distributor Lead
 - Retailer Lead
 - Doctor Lead
 - Hospital Lead
 - Franchise Lead
-

Error Handling

Scenario	Handling
Duplicate lead	Warning message
Missing contact details	Validation error
Territory not mapped	Restrict assignment

Acceptance Criteria

- Leads should create successfully.
 - Assignment should work correctly.
 - Notifications should trigger.
 - Duplicate detection should work.
-

Related Modules

- Notifications
 - MR Module
 - Reporting
 - Dashboard
-

Impact on Other Screens

- Lead funnel updated.
 - MR schedules updated.
 - CRM reports updated.
-

Module Communication

Source Module	Target Module	Communication
CRM	MR Module	Follow-up assignment
CRM	Notifications	Reminder alerts
CRM	Dashboard	Lead analytics

FR-2: Follow-Up & Meeting Management

User Story

As sales team, I want follow-up and meeting tracking so that lead conversion can be managed efficiently.

Flow of Work

1. Follow-up scheduled.
 2. Reminder generated.
 3. Meeting conducted.
 4. Notes captured.
 5. Next follow-up scheduled.
 6. Lead stage updated.
-

Error Handling

Scenario	Handling
Missed follow-up	Escalation alert
Meeting overlap	Schedule warning
Missing notes	Validation warning

Acceptance Criteria

- Follow-up reminders should trigger.
 - Meeting logs should save correctly.
 - Lead stages should update correctly.
-

Related Modules

- Notifications

- MR Module
 - Dashboard
 - Reporting
-

Impact on Other Screens

- CRM analytics updated.
 - MR activities updated.
 - Management visibility updated.
-

Module Communication

Source Module	Target Module	Communication
CRM	Notifications	Meeting reminders
CRM	MR Module	Follow-up tasks
CRM	Dashboard	Pipeline metrics

FR-3: Lead Conversion & Onboarding

User Story

As management, I want lead conversion tracking so that onboarding and revenue opportunities can be monitored.

Flow of Work

1. Lead qualified.
 2. Approval initiated.
 3. KYC/documents validated.
 4. Account created.
 5. Pricing/territory mapped.
 6. Distributor/Retailer activated.
-

Error Handling

Scenario	Handling
Missing KYC	Restrict activation
Duplicate onboarding	Validation warning
Territory conflict	Escalation alert

Acceptance Criteria

- Conversion workflow should complete correctly.
 - New account should activate successfully.
 - Territory mapping should work correctly.
-

Related Modules

- Distributor Portal
 - Retailer Module
 - Notifications
 - Accounting
-

Impact on Other Screens

- Distributor hierarchy updated.
 - CRM conversion reports updated.
 - Sales analytics updated.
-

Module Communication

Source Module	Target Module	Communication
CRM	Distributor Module	Account creation
CRM	Accounting	Ledger creation
CRM	Notifications	Onboarding alerts

9.6 QA VALIDATION CHECKLIST

- Verify lead creation.
- Verify duplicate lead detection.
- Verify follow-up scheduling.
- Verify meeting reminders.
- Verify lead conversion workflow.
- Verify onboarding workflow.
- Verify CRM analytics.
- Verify notification triggers.

MODULE 10 — ACCOUNTING & FINANCE MODULE

10.1 MODULE OVERVIEW

Module Name

Accounting & Finance Module

Objective

Manage complete financial operations for the Pharma ERP ecosystem including:

- Ledger management
- Outstanding tracking
- Payment management
- Profit & Loss
- Balance sheet
- GST reporting
- Commission calculations
- Bank reconciliation

This module acts as the centralized finance engine for all operational modules.

10.2 WHO WILL USE THIS MODULE

User Role	Purpose
Accounts Team	Financial operations
Finance Head	Financial monitoring
Management	Business profitability monitoring
Billing Team	Invoice/payment tracking
Auditors	Financial verification

10.3 PRE-CONDITIONS

- Billing module should be active.
 - Distributor/customer master should exist.
 - GST configuration should exist.
 - Ledger structure should be configured.
 - User permissions should be assigned.
-

10.4 POST-CONDITIONS

- Financial records updated.
 - Outstanding updated.
 - Reports generated.
 - Payment tracking updated.
 - Accounting entries maintained.
-

10.5 FUNCTIONAL REQUIREMENTS

FR-1: Ledger Management

User Story

As accounts team, I want ledger-wise transaction management so that all financial activities can be tracked accurately.

Flow of Work

1. Invoice generated.
 2. Ledger entry created.
 3. Payment adjustment processed.
 4. Outstanding recalculated.
 5. Transaction history maintained.
-

Error Handling

Scenario	Handling
Duplicate ledger entry	Restrict duplication
Missing customer mapping	Validation error
Invalid transaction amount	Validation restriction
Ledger imbalance	Reconciliation alert

Acceptance Criteria

- Ledger entries should generate correctly.
 - Outstanding should calculate correctly.
 - Payment adjustments should reflect accurately.
 - Ledger history should remain traceable.
-

Related Modules

- Billing
- Distributor Portal
- Retailer Orders
- Dashboard
- Notifications

Impact on Other Screens

- Outstanding visible in dashboards.
 - Distributor financial visibility updated.
 - Collection reports updated.
-

Module Communication

Source Module	Target Module	Communication
Billing	Accounting	Invoice sync
Payments	Accounting	Payment adjustment
Notifications	Accounting	Due alerts

FR-2: Outstanding & Aging Management

User Story

As finance management, I want aging analysis and outstanding tracking so that collections can be managed effectively.

Flow of Work

1. Outstanding generated from invoices.
 2. Due date monitored.
 3. Aging buckets calculated.
 4. Reminder notifications triggered.
 5. Collection updates reflected.
-

Aging Buckets

- 0–30 Days
 - 31–60 Days
 - 61–90 Days
 - 90+ Days
-

Error Handling

Scenario	Handling
Missing due date	Restrict invoice processing
Invalid payment update	Reconciliation validation
Outstanding mismatch	Audit alert

Acceptance Criteria

- Aging reports should calculate correctly.
 - Payment reminders should trigger.
 - Outstanding visibility should remain accurate.
-

Related Modules

- Billing
 - Notifications
 - Dashboard
 - Distributor Portal
-

Impact on Other Screens

- Dashboard financial KPIs updated.
 - Payment reminders visible.
 - Collection reports updated.
-

Module Communication

Source Module	Target Module	Communication
Accounting	Notifications	Due reminders
Billing	Accounting	Outstanding creation
Dashboard	Accounting	Financial analytics

FR-3: GST Reporting & Financial Statements

User Story

As finance management, I want GST and financial reporting so that compliance and profitability can be monitored.

Flow of Work

1. Transactions aggregated.
 2. GST calculations processed.
 3. Financial reports generated.
 4. Export/download enabled.
-

Reports Covered

- GST Reports
 - Sales Register
 - Purchase Register
 - Profit & Loss
 - Balance Sheet
 - Outstanding Reports
-

Error Handling

Scenario	Handling
GST mismatch	Validation alert
Incomplete transaction	Report exclusion warning
Missing accounting entries	Audit alert

Acceptance Criteria

- GST reports should calculate correctly.
 - Financial statements should balance correctly.
 - Reports should export successfully.
-

Related Modules

- Billing
 - Inventory
 - Dashboard
 - Reports
-

Impact on Other Screens

- Management dashboard updated.
 - Audit visibility enabled.
 - Financial KPIs updated.
-

Module Communication

Source Module	Target Module	Communication
Billing	Accounting	GST calculations
Inventory	Accounting	Stock valuation
Dashboard	Accounting	Financial KPIs

10.6 QA VALIDATION CHECKLIST

- Verify ledger creation.
 - Verify payment adjustments.
 - Verify aging reports.
 - Verify GST calculations.
 - Verify P&L calculations.
 - Verify balance sheet reports.
 - Verify reconciliation workflows.
 - Verify report exports.
-
-

MODULE 11 — ALERTS & NOTIFICATION ENGINE

11.1 MODULE OVERVIEW

Module Name

Alerts & Notification Engine

Objective

Provide centralized alerting and notification management across the ERP ecosystem.

This module manages:

- Payment reminders
 - Meeting reminders
 - Follow-up reminders
 - Expiry alerts
 - Reorder alerts
 - Dispatch alerts
 - CRM alerts
 - System notifications
-

11.2 WHO WILL USE THIS MODULE

User Role	Purpose
Management	Operational visibility
Sales Team	Follow-up reminders
MR Team	Schedule reminders
Accounts Team	Payment reminders
Warehouse Team	Dispatch alerts

11.3 PRE-CONDITIONS

- Notification engine configured.
 - User roles configured.
 - Event triggers configured.
 - Mobile/email/SMS channels configured.
-

11.4 POST-CONDITIONS

- Notifications triggered.
 - Alerts logged.
 - Reminder schedules maintained.
 - Escalations initiated.
-

11.5 FUNCTIONAL REQUIREMENTS

FR-1: Payment Reminder Notifications

User Story

As accounts management, I want automated payment reminders so that outstanding collections can improve.

Flow of Work

1. Due invoices monitored.
 2. Aging calculated.
 3. Reminder conditions matched.
 4. Notification triggered.
 5. Reminder logs maintained.
-

Notification Channels

- In-App
 - Push Notification
 - SMS Ready
 - WhatsApp Ready
 - Email Ready
-

Error Handling

Scenario	Handling
Notification failure	Retry mechanism
Invalid contact details	Validation alert
Duplicate reminders	Prevent duplicate alerts

Acceptance Criteria

- Reminders should trigger correctly.
 - Reminder logs should maintain history.
 - Notifications should reach intended users.
-

Related Modules

- Accounting
 - CRM
 - Dashboard
 - Distributor Portal
-

Impact on Other Screens

- Payment alerts visible in dashboard.
 - Outstanding alerts visible in distributor portal.
-

Module Communication

Source Module	Target Module	Communication
Accounting	Notifications	Due alerts
CRM	Notifications	Follow-up reminders
Dashboard	Notifications	Alert visibility

FR-2: Meeting & Follow-Up Reminders

User Story

As sales/MR team, I want automated reminders for meetings and follow-ups so that no opportunity is missed.

Flow of Work

1. Follow-up created.
 2. Reminder scheduled.
 3. Reminder engine triggered.
 4. Notification sent.
 5. Activity logs updated.
-

Error Handling

Scenario	Handling
Missed schedule	Escalation alert
Notification failure	Retry mechanism
Invalid schedule	Validation restriction

Acceptance Criteria

- Reminders should trigger on time.
 - Escalations should work correctly.
 - Follow-up logs should maintain history.
-

Related Modules

- CRM
 - MR Module
 - Dashboard
 - Reporting
-

Impact on Other Screens

- CRM activities updated.
 - MR dashboard reminders updated.
 - Lead funnel visibility updated.
-

Module Communication

Source Module	Target Module	Communication
CRM	Notifications	Follow-up alerts
MR Module	Notifications	Meeting reminders
Notifications	Dashboard	Reminder display

FR-3: Inventory & Dispatch Alerts

User Story

As operations management, I want inventory and dispatch alerts so that stock and logistics issues can be handled proactively.

Flow of Work

1. Inventory thresholds monitored.
 2. Dispatch statuses monitored.
 3. Alert conditions matched.
 4. Notifications generated.
 5. Escalation initiated if unresolved.
-

Alert Types

- Low Stock
 - Near Expiry
 - Pending Dispatch
 - Delayed Dispatch
 - Territory Exceptions
-

Error Handling

Scenario	Handling
Alert engine failure	Retry mechanism
Duplicate alerts	Restrict duplicate notifications
Invalid threshold	Default threshold applied

Acceptance Criteria

- Alerts should trigger accurately.
- Notifications should display correctly.
- Escalations should work.

Related Modules

- Inventory
 - Warehouse
 - Dashboard
 - Distributor Orders
-

Impact on Other Screens

- Inventory dashboard updated.
 - Dispatch monitoring updated.
 - Management alerts updated.
-

Module Communication

Source Module	Target Module	Communication
Inventory	Notifications	Low stock alerts
Warehouse	Notifications	Dispatch alerts
Dashboard	Notifications	Alert visibility

11.6 QA VALIDATION CHECKLIST

- Verify payment reminders.
 - Verify follow-up reminders.
 - Verify meeting alerts.
 - Verify low stock alerts.
 - Verify expiry alerts.
 - Verify dispatch alerts.
 - Verify escalation workflows.
 - Verify notification logs.
-
-

MODULE 12 — REPORTING & ANALYTICS MODULE

12.1 MODULE OVERVIEW

Module Name

Reporting & Analytics Module

Objective

Provide centralized operational, financial, inventory, CRM, and sales analytics for management decision-making.

This module manages:

- Sales reports
 - Inventory reports
 - CRM analytics
 - MR performance reports
 - Financial reports
 - Distributor performance
 - Territory analytics
 - Product analytics
-

12.2 WHO WILL USE THIS MODULE

User Role	Purpose
Management	Business analytics
Sales Head	Sales performance monitoring
Finance Team	Financial reporting
Operations Team	Inventory analytics

12.3 PRE-CONDITIONS

- Operational modules should be active.
 - Transactional data should exist.
 - Reporting permissions configured.
 - Dashboard integration configured.
-

12.4 POST-CONDITIONS

- Reports generated.
 - KPIs updated.
 - Analytics available.
 - Export/download available.
-

12.5 FUNCTIONAL REQUIREMENTS

FR-1: Sales & Distributor Analytics

User Story

As management, I want sales and distributor analytics so that business growth and performance can be monitored.

Flow of Work

1. Sales data aggregated.
2. Distributor performance calculated.
3. Territory analytics generated.
4. KPI dashboard updated.

5. Export options enabled.
-

Reports Covered

- State-wise Sales
 - Distributor Performance
 - Top Products
 - Territory Sales
 - Pending Collections
-

Error Handling

Scenario	Handling
Missing data	Show partial report warning
Large dataset	Pagination/export optimization
API timeout	Retry mechanism

Acceptance Criteria

- Reports should generate correctly.
 - Filters should work correctly.
 - Export should work successfully.
-

Related Modules

- Billing
 - Distributor Portal
 - Dashboard
 - Accounting
-

Impact on Other Screens

- Dashboard KPIs updated.

- Management analytics updated.
-

Module Communication

Source Module	Target Module	Communication
Billing	Reports	Sales data
Accounting	Reports	Financial analytics
Distributor Module	Reports	Distributor KPIs

FR-2: CRM & MR Performance Analytics

User Story

As management, I want CRM and MR analytics so that lead conversion and field-force performance can be monitored.

Flow of Work

1. CRM data aggregated.
 2. MR activity logs processed.
 3. Lead conversion metrics calculated.
 4. Performance reports generated.
-

Reports Covered

- Lead Funnel
 - Conversion Ratio
 - MR Visit Reports
 - Follow-up Reports
 - Territory Performance
 - Order Booking Reports
-

Error Handling

Scenario	Handling
Missing activity data	Partial report warning
Incorrect mapping	Validation alert
Data sync failure	Retry processing

Acceptance Criteria

- CRM reports should generate correctly.
 - MR reports should calculate correctly.
 - Conversion metrics should remain accurate.
-

Related Modules

- CRM
 - MR Module
 - Dashboard
 - Notifications
-

Impact on Other Screens

- Lead funnel visible in dashboard.
 - Incentive calculations updated.
 - Territory performance updated.
-

Module Communication

Source Module	Target Module	Communication
CRM	Reports	Lead analytics
MR Module	Reports	Activity analytics
Dashboard	Reports	KPI rendering

FR-3: Inventory & Financial Analytics

User Story

As management, I want inventory and financial analytics so that operational and financial planning can improve.

Flow of Work

1. Inventory data aggregated.
 2. Financial data aggregated.
 3. KPI calculations executed.
 4. Analytics reports generated.
-

Reports Covered

- Low Stock Report
 - Near Expiry Report
 - Dead Stock Report
 - P&L Analytics
 - Outstanding Analytics
 - GST Summary
-

Error Handling

Scenario	Handling
Missing inventory data	Partial warning
Incorrect stock mapping	Audit alert
Financial mismatch	Reconciliation alert

Acceptance Criteria

- Inventory reports should calculate correctly.
- Financial analytics should remain accurate.
- Reports should export correctly.

Related Modules

- Inventory
 - Accounting
 - Dashboard
 - Notifications
-

Impact on Other Screens

- Dashboard inventory KPIs updated.
 - Financial visibility updated.
 - Procurement planning updated.
-

Module Communication

Source Module	Target Module	Communication
Inventory	Reports	Stock analytics
Accounting	Reports	Financial analytics
Notifications	Reports	Alert analytics

12.6 QA VALIDATION CHECKLIST

- Verify report generation.
- Verify filters.
- Verify exports.
- Verify KPI calculations.
- Verify CRM analytics.
- Verify MR analytics.
- Verify inventory analytics.
- Verify financial analytics.

MODULE 13 — USER ROLE & PERMISSION MANAGEMENT

13.1 MODULE OVERVIEW

Module Name

User Role & Permission Management

Objective

Manage secure access control and permission-based visibility across the ERP ecosystem.

This module manages:

- User creation
 - Role assignment
 - Permission mapping
 - Module access control
 - Territory-based access
 - Activity restrictions
 - Audit logs
-

13.2 WHO WILL USE THIS MODULE

User Role	Purpose
Super Admin	Full access management
HR/Admin Team	User creation & mapping
Management	Access monitoring
IT Team	Security management

13.3 PRE-CONDITIONS

- Authentication module active.
 - Role hierarchy configured.
 - Modules configured.
 - Territory structure configured.
-

13.4 POST-CONDITIONS

- User access configured.
 - Permissions enforced.
 - Audit logs updated.
 - Security restrictions applied.
-

13.5 FUNCTIONAL REQUIREMENTS

FR-1: User Creation & Role Assignment

User Story

As admin, I want to create users and assign roles so that system access can be controlled properly.

Flow of Work

1. Admin opens user management.
 2. User details entered.
 3. Role selected.
 4. Territory assigned.
 5. Permissions mapped.
 6. Account activated.
-

User Types

- Super Admin
 - Management
 - Accounts Team
 - MR
 - Distributor
 - Retailer
 - Warehouse Team
 - CRM Team
-

Error Handling

Scenario	Handling
Duplicate email/mobile	Restrict creation
Missing role	Validation error
Invalid territory mapping	Restrict save
Unauthorized admin access	Access denied

Acceptance Criteria

- Users should create successfully.
 - Roles should apply correctly.
 - Territory mapping should work correctly.
 - Login access should reflect permissions.
-

Related Modules

- Authentication
 - Dashboard
 - CRM
 - Inventory
 - Billing
-

Impact on Other Screens

- Menu visibility changes dynamically.

- Dashboard visibility changes.
 - Report access restrictions applied.
-

Module Communication

Source Module	Target Module	Communication
User Module	Authentication	Login permissions
Roles Module	Dashboard	Access visibility
Territory Module	CRM	Territory access

FR-2: Role-Based Module Access Control

User Story

As management, I want role-based access restrictions so that users can access only permitted modules.

Flow of Work

1. Role permissions configured.
 2. Modules assigned.
 3. Actions restricted:
 - View
 - Create
 - Edit
 - Delete
 4. User session updated.
 5. Access restrictions enforced.
-

Error Handling

Scenario	Handling
----------	----------

Unauthorized access	Access denied
Missing permissions	Restrict module
Invalid role mapping	Validation alert

Acceptance Criteria

- Restricted modules should remain inaccessible.
 - Permissions should apply dynamically.
 - Unauthorized APIs should block access.
-

Related Modules

- Authentication
 - APIs
 - Dashboard
 - Reporting
-

Impact on Other Screens

- Buttons/actions hidden dynamically.
 - Reports filtered based on access.
-

Module Communication

Source Module	Target Module	Communication
Roles Module	APIs	Access validation
Roles Module	Dashboard	UI restrictions
Roles Module	Reports	Data visibility

FR-3: Activity Logs & Audit Trail

User Story

As management, I want audit trails for user activities so that operational transparency and accountability can be maintained.

Flow of Work

1. User activity captured.
 2. Activity logs stored.
 3. Timestamp recorded.
 4. Device/IP captured.
 5. Audit reports generated.
-

Logged Activities

- Login/Logout
 - Invoice creation
 - Product edits
 - Inventory changes
 - CRM activities
 - Payment updates
-

Error Handling

Scenario	Handling
Log failure	Retry logging
Missing activity mapping	Alert admin
Storage limit issue	Archive logs

Acceptance Criteria

- Activities should log correctly.
 - Audit reports should generate.
 - Logs should remain tamper-resistant.
-

Related Modules

- Authentication
 - Billing
 - CRM
 - Inventory
 - Accounting
-

Impact on Other Screens

- Activity visibility in admin panel.
 - Audit reports visible in reporting module.
-

Module Communication

Source Module	Target Module	Communication
All Modules	Audit Logs	Activity tracking
Audit Logs	Reports	Audit analytics
Authentication	Audit Logs	Login tracking

13.6 QA VALIDATION CHECKLIST

- Verify user creation.
 - Verify role mapping.
 - Verify permission restrictions.
 - Verify API access restrictions.
 - Verify territory mapping.
 - Verify audit logs.
 - Verify activity tracking.
 - Verify access visibility.
-
-

MODULE 14 — DISPATCH & LOGISTICS MANAGEMENT

14.1 MODULE OVERVIEW

Module Name

Dispatch & Logistics Management

Objective

Manage product dispatch and logistics operations across warehouses, distributors, and retailers.

This module manages:

- Dispatch planning
 - Shipment processing
 - LR tracking
 - Transport challan
 - Delivery tracking
 - Dispatch status updates
 - Pending dispatch monitoring
-

14.2 WHO WILL USE THIS MODULE

User Role	Purpose
Warehouse Team	Dispatch execution
Logistics Team	Shipment coordination
Operations Team	Dispatch monitoring
Management	Logistics visibility
Distributor	Delivery tracking

14.3 PRE-CONDITIONS

- Orders should exist.
 - Inventory should be available.
 - Warehouse configured.
 - Transport setup configured.
 - Customer/distributor mapping completed.
-

14.4 POST-CONDITIONS

- Dispatch generated.
 - Shipment tracked.
 - Stock reduced.
 - Delivery status updated.
 - Notifications triggered.
-

14.5 FUNCTIONAL REQUIREMENTS

FR-1: Dispatch Planning & Processing

User Story

As warehouse team, I want to process dispatches so that approved orders can be delivered efficiently.

Flow of Work

1. Approved orders fetched.
2. Inventory validated.
3. Dispatch created.
4. Batch allocation finalized.

5. Challan generated.
 6. Shipment assigned.
-

Error Handling

Scenario	Handling
Insufficient stock	Restrict dispatch
Invalid batch	Validation error
Duplicate dispatch	Warning restriction
Missing transport details	Validation alert

Acceptance Criteria

- Dispatch should generate correctly.
 - Inventory should reduce correctly.
 - Batch allocation should remain accurate.
 - Challan generation should work.
-

Related Modules

- Inventory
 - Billing
 - Distributor Orders
 - Warehouse
-

Impact on Other Screens

- Dispatch status updated in distributor portal.
 - Inventory dashboard updated.
 - Pending dispatch analytics updated.
-

Module Communication

Source Module	Target Module	Communication
Orders	Dispatch	Approved orders
Inventory	Dispatch	Stock validation
Dispatch	Billing	Invoice linkage

FR-2: LR & Shipment Tracking

User Story

As logistics team, I want shipment tracking and LR management so that delivery movement can be monitored.

Flow of Work

1. LR number generated/added.
 2. Transport assigned.
 3. Shipment status updated.
 4. Delivery tracking maintained.
 5. Delivery completion marked.
-

Shipment Statuses

- Pending
 - Packed
 - Dispatched
 - In Transit
 - Delivered
 - Failed Delivery
-

Error Handling

Scenario	Handling
----------	----------

Invalid LR number	Validation error
Shipment update failure	Retry mechanism
Delivery mismatch	Escalation alert

Acceptance Criteria

- Shipment statuses should update correctly.
 - LR tracking should work correctly.
 - Delivery logs should maintain correctly.
-

Related Modules

- Notifications
 - Distributor Portal
 - Reporting
 - Dashboard
-

Impact on Other Screens

- Distributor dispatch tracking updated.
 - Logistics analytics updated.
 - Pending delivery alerts updated.
-

Module Communication

Source Module	Target Module	Communication
Dispatch	Distributor Portal	Shipment updates
Logistics	Notifications	Delivery alerts
Dispatch	Reports	Logistics analytics

FR-3: Pending Dispatch & Delivery Alerts

User Story

As operations management, I want dispatch delay alerts so that logistics bottlenecks can be resolved quickly.

Flow of Work

1. Pending dispatch monitored.
 2. SLA timelines checked.
 3. Delay conditions matched.
 4. Alerts triggered.
 5. Escalation initiated.
-

Error Handling

Scenario	Handling
Missing dispatch status	Validation alert
Duplicate alerts	Restrict duplication
Escalation failure	Retry escalation

Acceptance Criteria

- Delay alerts should trigger correctly.
 - Escalations should work properly.
 - Pending dispatch reports should remain accurate.
-

Related Modules

- Notifications
- Dashboard
- Reporting
- Distributor Portal

Impact on Other Screens

- Dashboard alerts updated.
 - Distributor visibility updated.
 - Logistics KPIs updated.
-

Module Communication

Source Module	Target Module	Communication
Dispatch	Notifications	Delay alerts
Notifications	Dashboard	Alert visibility
Dispatch	Reports	Pending dispatch analytics

14.6 QA VALIDATION CHECKLIST

- Verify dispatch generation.
 - Verify batch allocation.
 - Verify challan generation.
 - Verify shipment tracking.
 - Verify LR tracking.
 - Verify delivery status updates.
 - Verify delay alerts.
 - Verify logistics reports.
-
-

MODULE 15 — MOBILE APPLICATION MODULE

15.1 MODULE OVERVIEW

Module Name

Mobile Application Module

Objective

Provide mobile access to different stakeholders for operational accessibility and real-time business visibility.

Mobile applications covered:

- MR App
 - Retailer App
 - Distributor App
 - Management Dashboard App
-

15.2 WHO WILL USE THIS MODULE

User Role	Purpose
MR Team	Field operations
Retailers	Product ordering
Distributors	Order & finance visibility
Management	Business monitoring

15.3 PRE-CONDITIONS

- User accounts should exist.
 - Mobile APIs should be active.
 - Push notification setup completed.
 - GPS permissions configured.
 - Internet connectivity available.
-

15.4 POST-CONDITIONS

- Mobile activities synced.
 - Orders processed.
 - Notifications delivered.
 - Reports updated.
 - GPS activities tracked.
-

15.5 FUNCTIONAL REQUIREMENTS

FR-1: Mobile Authentication & Session Management

User Story

As a mobile user, I want secure mobile login so that my business activities remain secure.

Flow of Work

1. User opens app.
 2. Credentials entered.
 3. Authentication validated.
 4. Session token generated.
 5. Dashboard loads.
-

Error Handling

Scenario	Handling
Invalid login	Validation error
Session expired	Auto logout
Network unavailable	Retry/login cache

Unauthorized device Security alert

Acceptance Criteria

- Login should work correctly.
 - Sessions should remain secure.
 - Auto logout should work properly.
 - Token refresh should work correctly.
-

Related Modules

- Authentication
 - Notifications
 - Dashboard
 - CRM
-

Impact on Other Screens

- Mobile dashboard visibility updated.
 - Push notifications enabled.
-

Module Communication

Source Module	Target Module	Communication
Authentication	Mobile App	Session validation
Notifications	Mobile App	Push alerts
Dashboard	Mobile App	KPI visibility

FR-2: Mobile Order Management

User Story

As retailer/distributor/MR, I want mobile order placement so that procurement and sales operations become faster.

Flow of Work

1. Product catalog loaded.
 2. Product selected.
 3. Quantity validated.
 4. Pricing/schemes applied.
 5. Order submitted.
 6. Dispatch workflow initiated.
-

Error Handling

Scenario	Handling
Product unavailable	Restrict ordering
Invalid pricing	Refresh pricing
Internet failure	Offline retry queue
Duplicate order	Warning message

Acceptance Criteria

- Orders should sync correctly.
 - Pricing should apply correctly.
 - Offline retry should work.
 - Order status should update correctly.
-

Related Modules

- Inventory
- Billing
- Distributor Orders
- Notifications

Impact on Other Screens

- Dispatch queue updated.
 - Inventory analytics updated.
 - Mobile dashboards updated.
-

Module Communication

Source Module	Target Module	Communication
Mobile App	Orders	Order sync
Inventory	Mobile App	Product visibility
Notifications	Mobile App	Order alerts

FR-3: Mobile Notifications & Real-Time Alerts

User Story

As a mobile user, I want real-time alerts so that I can respond quickly to operational activities.

Flow of Work

1. Notification triggered.
 2. Push notification generated.
 3. Mobile delivery attempted.
 4. Notification displayed.
 5. User action logged.
-

Notification Types

- Payment reminders
 - Dispatch updates
 - Meeting reminders
 - Follow-up reminders
 - Low stock alerts
 - CRM activities
-

Error Handling

Scenario	Handling
Push failure	Retry mechanism
Device offline	Delayed delivery
Notification duplication	Restrict duplicates

Acceptance Criteria

- Notifications should deliver correctly.
 - Alerts should open related screens.
 - Notification history should maintain correctly.
-

Related Modules

- Notifications
 - CRM
 - Dispatch
 - Inventory
 - Accounting
-

Impact on Other Screens

- Real-time dashboard updates.
 - User activity logs updated.
-

Module Communication

Source Module	Target Module	Communication
Notifications	Mobile App	Push delivery
CRM	Mobile App	Follow-up alerts
Dispatch	Mobile App	Shipment updates

15.6 QA VALIDATION CHECKLIST

- Verify mobile login.
- Verify session handling.
- Verify mobile order placement.
- Verify push notifications.
- Verify offline retry handling.
- Verify dashboard loading.
- Verify GPS permissions.
- Verify real-time alerts.

MODULE 16 — EXPORT & GLOBAL OPERATIONS MODULE

16.1 MODULE OVERVIEW

Module Name

Export & Global Operations Module

Objective

Support international pharma operations and export workflows.

This module manages:

- Multi-currency support
 - Export invoicing
 - Country-wise pricing
 - IEC workflows
 - HS code mapping
 - Export documentation
 - Shipping documentation
 - FDA/GMP document management
-

16.2 WHO WILL USE THIS MODULE

User Role	Purpose
Export Team	Export processing
Finance Team	Export billing
Management	Global business visibility
Logistics Team	Shipment coordination

16.3 PRE-CONDITIONS

- Export setup configured.
 - Currency master configured.
 - Country pricing configured.
 - Export customer profiles created.
 - Shipping workflows configured.
-

16.4 POST-CONDITIONS

- Export orders processed.
 - Export invoices generated.
 - Currency calculations completed.
 - Shipping documentation maintained.
-

16.5 FUNCTIONAL REQUIREMENTS

FR-1: Multi-Currency Pricing & Export Orders

User Story

As export team, I want multi-currency order handling so that international orders can be processed efficiently.

Flow of Work

1. Export customer selected.
 2. Country pricing fetched.
 3. Currency conversion applied.
 4. Export order created.
 5. Export invoice generated.
-

Error Handling

Scenario	Handling
Missing exchange rate	Validation restriction
Invalid export customer	Restrict processing
Currency mismatch	Validation alert

Acceptance Criteria

- Currency calculations should work correctly.
 - Export orders should save successfully.
 - Country pricing should apply correctly.
-

Related Modules

- Billing
 - Accounting
 - Inventory
 - Logistics
-

Impact on Other Screens

- Export analytics updated.
 - Financial reports updated.
 - Inventory allocations updated.
-

Module Communication

Source Module	Target Module	Communication
Currency Module	Export Orders	Currency conversion
Billing	Accounting	Export invoice sync
Inventory	Export Orders	Stock allocation

FR-2: Export Documentation Management

User Story

As export operations team, I want export document management so that shipment compliance can be maintained.

Flow of Work

1. Export shipment created.
2. Documents uploaded/generated.
3. IEC/HS code mapped.

4. Compliance validation executed.
 5. Shipment approved.
-

Documents Covered

- Export Invoice
 - Packing List
 - Shipping Documents
 - FDA Documents
 - GMP Certificates
 - IEC Documents
-

Error Handling

Scenario	Handling
Missing documents	Restrict approval
Invalid HS code	Validation error
Document upload failure	Retry upload

Acceptance Criteria

- Documents should upload correctly.
 - Compliance validation should work.
 - Shipment approval should depend on documentation.
-

Related Modules

- Logistics
 - Accounting
 - Reporting
 - Dashboard
-

Impact on Other Screens

- Export shipment visibility updated.
 - Compliance reports updated.
-

Module Communication

Source Module	Target Module	Communication
Export Module	Logistics	Shipment processing
Export Module	Reports	Export analytics
Export Module	Dashboard	Export KPIs

FR-3: Country-Wise Pricing & Taxation

User Story

As management, I want country-wise pricing and taxation so that global operations can be controlled accurately.

Flow of Work

1. Country selected.
 2. Country pricing fetched.
 3. Tax rules applied.
 4. Export invoice calculated.
 5. Reports updated.
-

Error Handling

Scenario	Handling
Missing country pricing	Validation restriction
Invalid taxation rule	Validation error

Duplicate pricing setup Warning alert

Acceptance Criteria

- Country pricing should apply correctly.
 - Export taxation should calculate correctly.
 - Pricing reports should update correctly.
-

Related Modules

- Accounting
 - Billing
 - Reporting
 - Dashboard
-

Impact on Other Screens

- Export profitability updated.
 - Global sales analytics updated.
-

Module Communication

Source Module	Target Module	Communication
Pricing Module	Export Billing	Country pricing
Accounting	Export Reports	Export taxation
Dashboard	Export Analytics	Global KPIs

16.6 QA VALIDATION CHECKLIST

- Verify multi-currency calculations.
- Verify export invoices.
- Verify document uploads.

- Verify HS code mapping.
 - Verify export reports.
 - Verify country pricing.
 - Verify export taxation.
-
-

MODULE 17 — AI ANALYTICS & SMART AUTOMATION MODULE

17.1 MODULE OVERVIEW

Module Name

AI Analytics & Smart Automation Module

Objective

Provide intelligent analytics and automated operational insights.

This module manages:

- Sales trend analytics
 - Product performance insights
 - Smart reorder suggestions
 - Predictive inventory alerts
 - Smart follow-up reminders
 - AI-driven dashboards
-

17.2 WHO WILL USE THIS MODULE

User Role	Purpose
Management	Business intelligence
Sales Team	Sales insights

Inventory Team Smart inventory planning

CRM Team Follow-up optimization

17.3 PRE-CONDITIONS

- Transactional data should exist.
 - Reporting module active.
 - Inventory data available.
 - CRM activities available.
-

17.4 POST-CONDITIONS

- AI insights generated.
 - Recommendations displayed.
 - Predictive alerts triggered.
 - Smart analytics updated.
-

17.5 FUNCTIONAL REQUIREMENTS

FR-1: Product Performance Analytics

User Story

As management, I want AI-driven product analytics so that fast-moving and slow-moving products can be identified.

Flow of Work

1. Sales data aggregated.
2. Product movement analyzed.

3. AI engine calculates trends.
 4. Recommendations generated.
 5. Dashboard updated.
-

Insights Covered

- Fast Moving Products
 - Slow Moving Products
 - Dead Stock
 - Territory-wise Product Demand
 - Seasonal Trends
-

Error Handling

Scenario	Handling
Missing analytics data	Partial insights warning
Data sync issue	Retry processing
Invalid product mapping	Validation alert

Acceptance Criteria

- Analytics should calculate correctly.
 - Trends should display accurately.
 - Recommendations should refresh dynamically.
-

Related Modules

- Inventory
 - Reporting
 - Dashboard
 - Billing
-

Impact on Other Screens

- Smart insights visible in dashboard.
- Procurement planning updated.

Module Communication

Source Module	Target Module	Communication
Billing	AI Engine	Sales analytics
Inventory	AI Engine	Stock trends
AI Engine	Dashboard	Recommendations

FR-2: Smart Reorder & Inventory Prediction

User Story

As inventory management, I want smart reorder suggestions so that stock shortages can be minimized.

Flow of Work

1. Inventory levels monitored.
 2. Consumption patterns analyzed.
 3. AI prediction engine executed.
 4. Reorder suggestions generated.
 5. Alerts triggered.
-

Error Handling

Scenario	Handling
Missing inventory data	Restrict prediction

Incorrect consumption mapping Validation warning

Prediction engine timeout Retry process

Acceptance Criteria

- Smart reorder suggestions should generate.
 - Inventory prediction should remain accurate.
 - Alerts should trigger correctly.
-

Related Modules

- Inventory
 - Notifications
 - Dashboard
 - Procurement
-

Impact on Other Screens

- Low stock analytics updated.
 - Procurement recommendations updated.
-

Module Communication

Source Module	Target Module	Communication
Inventory	AI Engine	Consumption data
AI Engine	Notifications	Smart alerts
AI Engine	Dashboard	Recommendations

FR-3: Smart CRM Follow-Up Recommendations

User Story

As CRM/sales team, I want AI-driven follow-up recommendations so that lead conversion can improve.

Flow of Work

1. CRM activities analyzed.
 2. Lead engagement patterns processed.
 3. AI engine predicts priority leads.
 4. Follow-up recommendations generated.
 5. Reminder suggestions displayed.
-

Error Handling

Scenario	Handling
Missing CRM activity	Partial recommendation
Invalid lead mapping	Validation warning
AI engine failure	Fallback logic

Acceptance Criteria

- Lead prioritization should work correctly.
 - Follow-up recommendations should display.
 - CRM productivity insights should update.
-

Related Modules

- CRM
- Notifications

- Dashboard
 - Reporting
-

Impact on Other Screens

- Lead funnel analytics updated.
 - CRM dashboards updated.
-

Module Communication

Source Module	Target Module	Communication
CRM	AI Engine	Lead activity data
AI Engine	Notifications	Smart reminders
AI Engine	Dashboard	CRM insights

17.6 QA VALIDATION CHECKLIST

- Verify analytics generation.
 - Verify reorder predictions.
 - Verify AI recommendations.
 - Verify dashboard insights.
 - Verify smart alerts.
 - Verify CRM prioritization.
 - Verify analytics refresh.
-
-

MODULE 18 — SYSTEM ADMINISTRATION, BACKUP & CLOUD INFRASTRUCTURE MODULE

18.1 MODULE OVERVIEW

Module Name

System Administration, Backup & Cloud Infrastructure Module

Objective

Manage infrastructure, deployment, security, backups, monitoring, and production environment operations.

This module manages:

- Cloud hosting
 - Backup management
 - Security configurations
 - Environment setup
 - Monitoring
 - Server health
 - Disaster recovery
 - API monitoring
-

18.2 WHO WILL USE THIS MODULE

User Role	Purpose
DevOps Team	Infrastructure management
IT Team	System administration
Management	Uptime visibility
Security Team	Security monitoring

18.3 PRE-CONDITIONS

- Production infrastructure configured.
- Domain setup completed.
- SSL configured.

- Database deployed.
 - Monitoring tools configured.
-

18.4 POST-CONDITIONS

- System deployed successfully.
 - Backups maintained.
 - Monitoring active.
 - Security policies enforced.
 - Recovery systems enabled.
-

18.5 FUNCTIONAL REQUIREMENTS

FR-1: Cloud Deployment & Environment Management

User Story

As DevOps team, I want secure cloud deployment so that the ERP system remains scalable and accessible across India.

Flow of Work

1. Server provisioned.
 2. Environment configured.
 3. Database deployed.
 4. APIs deployed.
 5. SSL configured.
 6. Monitoring enabled.
-

Error Handling

Scenario	Handling
Deployment failure	Rollback deployment
SSL failure	Restrict production release
Database migration failure	Restore backup

Acceptance Criteria

- Production deployment should work correctly.
 - SSL should remain active.
 - APIs should remain accessible.
 - Monitoring should function correctly.
-

Related Modules

- APIs
 - Authentication
 - Mobile Apps
 - Notifications
-

Impact on Other Screens

- System availability maintained.
 - Mobile/API accessibility maintained.
-

Module Communication

Source Module	Target Module	Communication
Infrastructure	APIs	Hosting environment
Infrastructure	Database	Connectivity
Monitoring	Dashboard	Server health

FR-2: Backup & Disaster Recovery

User Story

As management, I want automated backup and recovery systems so that business continuity remains protected.

Flow of Work

1. Backup scheduler triggered.
 2. Database backup created.
 3. File backup created.
 4. Backup stored securely.
 5. Recovery checkpoints maintained.
-

Error Handling

Scenario	Handling
Backup failure	Retry backup
Storage full	Escalation alert
Recovery corruption	Restore fallback backup

Acceptance Criteria

- Scheduled backups should run correctly.
 - Recovery should work successfully.
 - Backup logs should maintain history.
-

Related Modules

- Database
- File Storage
- Monitoring
- Notifications

Impact on Other Screens

- System reliability improved.
- Downtime risks minimized.

Module Communication

Source Module	Target Module	Communication
Backup Engine	Database	Backup operations
Backup Engine	Notifications	Backup alerts
Recovery Engine	Infrastructure	Recovery workflows

FR-3: Security Monitoring & API Health Monitoring

User Story

As IT/security team, I want security and API monitoring so that system reliability and protection remain strong.

Flow of Work

1. API monitoring initiated.
2. Security events tracked.
3. Failed login attempts monitored.
4. Server metrics monitored.
5. Alerts triggered for anomalies.

Error Handling

Scenario	Handling
----------	----------

API downtime	Escalation alert
Suspicious activity	Security restriction
High server load	Auto-scaling alert

Acceptance Criteria

- API monitoring should work correctly.
 - Security alerts should trigger.
 - Downtime alerts should function properly.
-

Related Modules

- Authentication
 - APIs
 - Dashboard
 - Notifications
-

Impact on Other Screens

- System health visibility updated.
 - Security alerts visible in admin dashboard.
-

Module Communication

Source Module	Target Module	Communication
Monitoring Engine	Dashboard	Health metrics
Security Engine	Notifications	Security alerts
APIs	Monitoring	API health data

18.6 QA VALIDATION CHECKLIST

- Verify deployment.

- Verify SSL configuration.
- Verify API availability.
- Verify backup scheduling.
- Verify disaster recovery.
- Verify security monitoring.
- Verify API monitoring.
- Verify uptime alerts.